1. Circular table with two people not adjacent (4 points)

We find this problem is a circular permutation. Therefore, there are 8 distinct friends sitting around a circular table, so we remember that arrangements are counted up to rotation.

Total circular seatings. Using the formula from class, we find the number of circular permutations of 8 distinct people is

$$(8-1)! = 7!.$$

Forbidden seatings (Cady next to Regina). We wish to treat the Cady and Regina rule as merging Cady and Regina into a single unit. Therefore, after the merge, we have 7 objects to arrange around the circle, giving

$$(7-1)! = 6!$$

circular arrangements of the 7 objects. Inside the unit, we find that Cady and Regina can be ordered in 2 ways, so the number of forbidden seatings is

$$2 \cdot 6!.$$

Allowed seatings. Therefore, we find the total number of allowed seatings to be defined as:

$$7! - 2 \cdot 6! = 6!(7-2) = 5 \cdot 6! = 3600.$$

2. Carousel: 5 riders, 2 operators (4 points)

We find this problem to be similar to circular permutations. There are 7 friends. Each round has 5 riders on the carousel (with 5 positions in a circle) and the remaining 2 are operators. The two operator roles are indistinguishable.

Choose the operators. We begin by choosing which 2 friends operate:

$$\binom{7}{2}.$$

Arrange the riders. Trivially, we find the remaining 5 friends must ride. We count circular permutations of the 5 riders:

$$(5-1)! = 4!.$$

Total. Therefore, we find the total number of seatings to be:

$$\binom{7}{2} \cdot 4! = 21 \cdot 24 = 504.$$

3. Permutations of `SENSELESSNESS` (4 points)

Using our fingers to count, we find the word `SENSELESSNESS` has 13 letters with the following repetitions:

$$S^6, \quad E^4, \quad N^2, \quad L^1.$$

Therefore, we find the number of permutations to be

$$\frac{13!}{6!\,4!\,2!}.$$

4. `COMBINATORIAL` with A's not consecutive (4 points)

Using a similar strategy to before, we find the word `COMBINATORIAL` has 13 letters with repeated letters:

$$A^2, \quad O^2, \quad I^2,$$

and all other letters occur once.

Total permutations (no restriction). Therefore, we find the following:

$$\frac{13!}{2!\,2!\,2!}.$$

Forbidden permutations (A's consecutive). If the two A's are consecutive, treat `AA` as one unit (similar to problem 1). Therefore, we find there are 12 objects total:

$$(\mathtt{AA}), C, M, B, N, T, R, L, \text{ and } O, O, I, I,$$

where only O and I are repeated (each twice). Relatively trivially, we then find the number of forbidden permutations is

$$\frac{12!}{2!\,2!}.$$

Allowed permutations. Finally, given the constraints, we find the number of allowed permutations to be:

$$\frac{13!}{2!\,2!\,2!} - \frac{12!}{2!\,2!}.$$

5. Combinatorial proof (4 points)

We wish to prove

$$26 \cdot 25 \cdot 10 \cdot 9 \cdot 8 \cdot \binom{5}{2} = \binom{26}{2}\binom{10}{3} 5!$$

LHS

We begin by identifying a method to produce the left hand side:

- Choose which 2 of the 5 positions are letters: $\binom{5}{2}$.
- Fill those two letter positions with ordered distinct letters: $26 \cdot 25$.
- Fill the remaining three positions with ordered distinct digits: $10 \cdot 9 \cdot 8$.

Multiplying (rule of product) gives

$$\binom{5}{2} \cdot (26 \cdot 25) \cdot (10 \cdot 9 \cdot 8),$$

which is exactly the left-hand side.

RHS

We then identify a method to product the right hand side:

- Choose the set of 2 letters (order not yet chosen): $\binom{26}{2}$.
- Choose the set of 3 digits (order not yet chosen): $\binom{10}{3}$.
- Now there are 5 distinct chosen symbols; arrange them in the 5 positions in any order: 5!.

Thus the number of plates is

$$\binom{26}{2}\binom{10}{3}5!,$$

which is the right-hand side.

Conclusion

Therefore, since both expressions count the same set of license plates, they are equivalent.